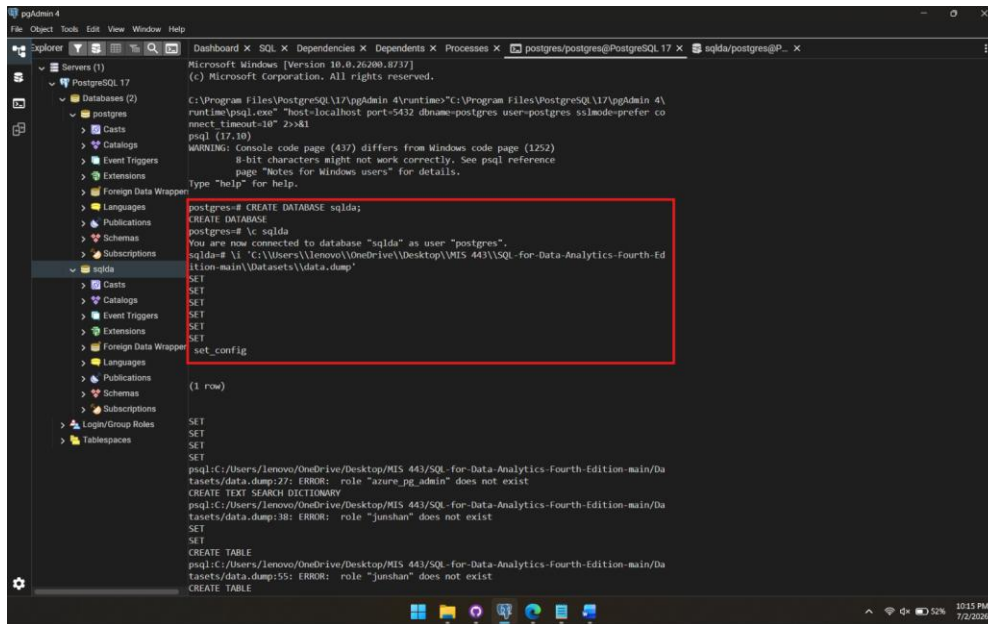


ASSIGNMENT 1

Basic SQL Install and Load Database with PostgreSQL

1. Installing PostgreSQL and Creating the sqlda Database

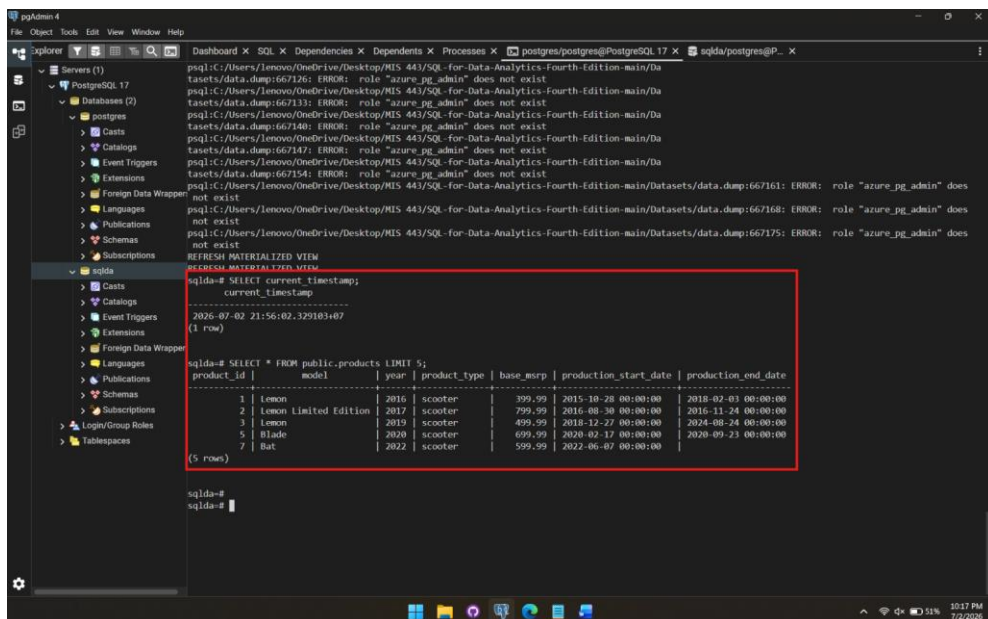
PostgreSQL version 17.10 was installed. The psql command-line tool was used to create a new database and import the sample dataset.



```
psql=# CREATE DATABASE sqlda;
CREATE DATABASE
psql=# \c sqlda
You are now connected to database "sqlda" as user "postgres".
sqlda=# \i 'C:\Users\lenovo\OneDrive\Desktop\MIS 443\SQL-for-Data-Analytics-Fourth-Edition-main\Datasets\data.dump'
SET
SET
SET
SET
SET
SET
set_config
(1 row)

SET
SET
SET
psql:C:\Users\lenovo\OneDrive\Desktop\MIS 443\SQL-for-Data-Analytics-Fourth-Edition-main\Datasets\data.dump:27: ERROR: role "azure_pg_admin" does not exist
CREATE TEXT SEARCH DICTIONARY
psql:C:\Users\lenovo\OneDrive\Desktop\MIS 443\SQL-for-Data-Analytics-Fourth-Edition-main\Datasets\data.dump:38: ERROR: role "junshan" does not exist
SET
CREATE TABLE
psql:C:\Users\lenovo\OneDrive\Desktop\MIS 443\SQL-for-Data-Analytics-Fourth-Edition-main\Datasets\data.dump:55: ERROR: role "junshan" does not exist
CREATE TABLE
```

The connection was then verified in pgAdmin:



```
sqlda=# SELECT current_timestamp;
current_timestamp
-----
2026-07-02 21:56:02.329103+07
(1 row)

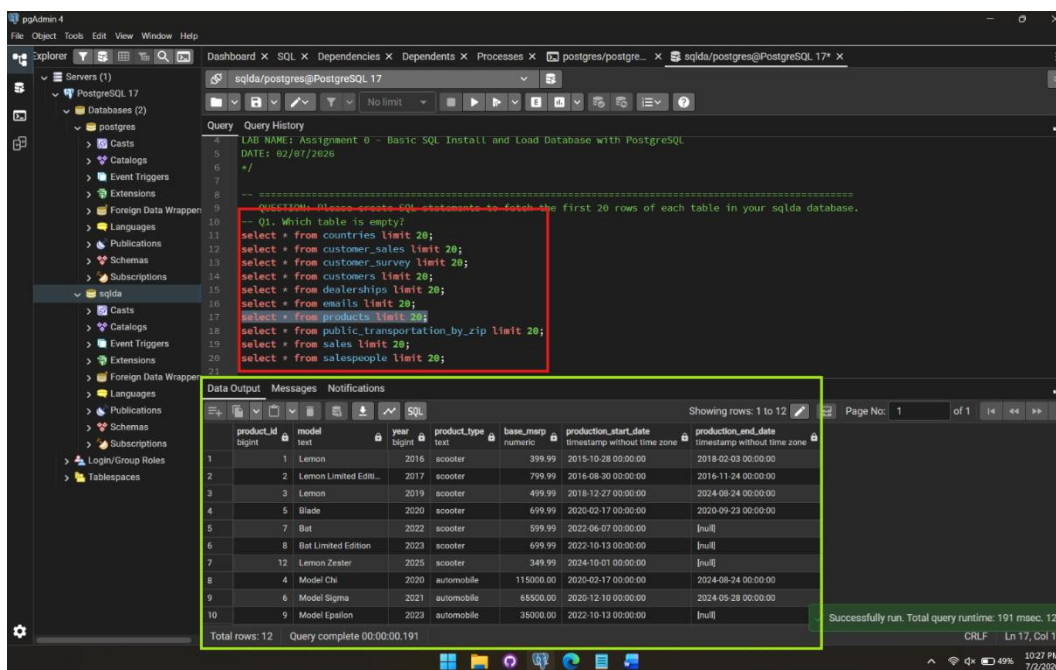
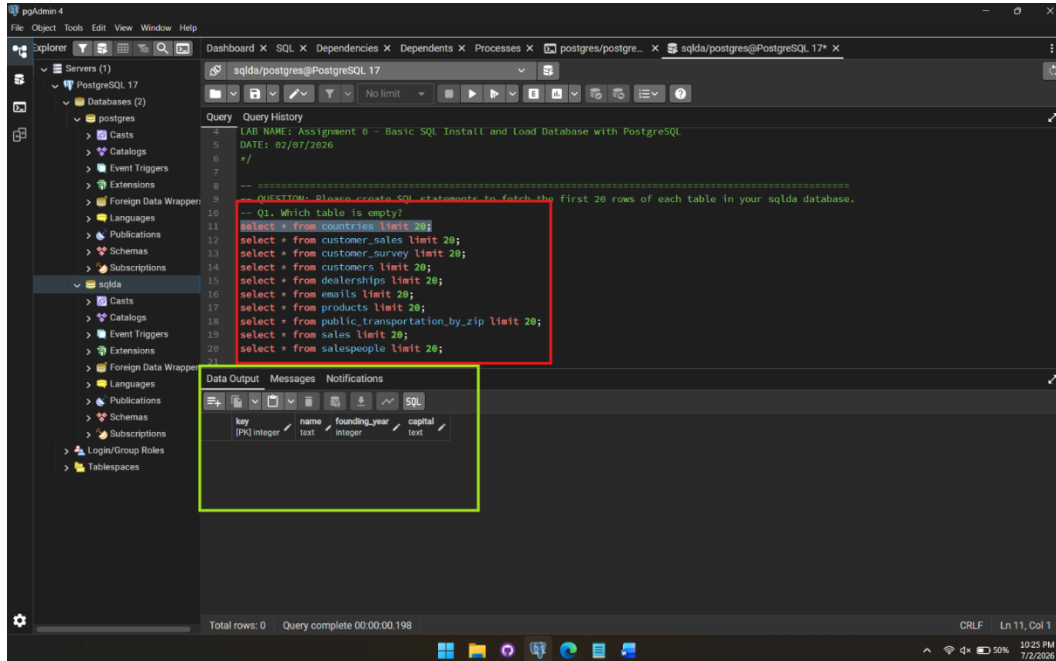
sqlda=# SELECT * FROM public.products LIMIT 5;
 product_id | model | year | product_type | base_msrp | production_start_date | production_end_date
-----
1 | Lemon | 2016 | scooter | 399.99 | 2015-10-28 00:00:00 | 2018-02-01 00:00:00
2 | Lemon Limited Edition | 2017 | scooter | 799.99 | 2016-08-30 00:00:00 | 2016-11-24 00:00:00
3 | Lemon | 2019 | scooter | 499.99 | 2018-12-27 00:00:00 | 2024-08-24 00:00:00
5 | Blade | 2020 | scooter | 699.99 | 2020-02-17 00:00:00 | 2020-09-23 00:00:00
7 | Bat | 2022 | scooter | 599.99 | 2022-06-07 00:00:00 | 
(5 rows)
```

2. Assignment Question

Create SQL statements to fetch the first 20 rows of each table in the *sqlda* database, then answer the following two questions.

QUESTION 1 Which table is empty?

METHOD First, run a SELECT ... LIMIT 20 query on every table to preview the data:



EXPLANATION

SELECT * retrieves every column of the table, FROM specifies which table to query, and LIMIT 20 caps the output at the first 20 rows so the result set stays readable.

ANSWER

The countries table is empty (0 rows).

QUESTION 2 Which table has fewer than 20 records?

METHOD Using COUNT(*) combined with UNION ALL to get the exact row count for every table.

The screenshot shows the pgAdmin 4 interface. The SQL editor contains a query that counts the number of rows in ten tables. The query is as follows:

```
-- Q2. Which table has fewer than 20 records?
select 'countries' as table_name, count(*) from countries
union all select 'customer_sales', count(*) from customer_sales
union all select 'customer_survey', count(*) from customer_survey
union all select 'customers', count(*) from customers
union all select 'dealerships', count(*) from dealerships
union all select 'emails', count(*) from emails
union all select 'products', count(*) from products
union all select 'public_transportation_by_zip', count(*) from public_transportation_by_zip
union all select 'sales', count(*) from sales
union all select 'salespeople', count(*) from salespeople;
```

The results are displayed in the Data Output tab, showing a table with two columns: table_name and count. The results are as follows:

table_name	count
products	12
dealerships	20
salespeople	300
countries	0
customer_survey	32
public_transportation_by...	15412
sales	37711
customers	50000
customer_sales	50000
emails	418158

Annotations in the image point to the 'products' and 'countries' rows, stating: "Both of these tables have fewer than 20 records."

EXPLANATION

"COUNT(*) returns the exact row count of a table. Each sub-query is labeled with a sub-name (table_name), so the result shows which table each count belongs to. UNION ALL then combines all ten counts into a single result set for easy comparison."

ANSWER

Two tables have fewer than 20 records: countries (0 rows) and products (12 rows).